

NAME: _____

PERIOD: _____

Two Measurements per Technician

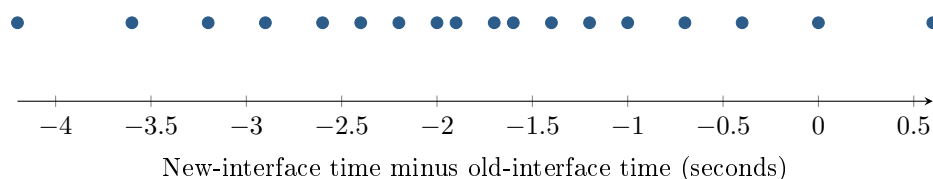
AP Statistics · Unit 4 · Topic 4.4

[STAR] THE PROBLEM

Suppose a company takes an SRS of 18 of its 300 technicians. Each completes the same task with old and new interfaces; interface order is randomly assigned. The plot shows $d = \text{new time} - \text{old time}$. Researchers ask whether the new interface reduces the population mean time.

Nia: “The technician seems to be the independent unit, so I would build one difference per person; the subtraction order matters for the tail.”

Colin: “There are 36 recorded times, so I would use two independent samples of 18 and a two-sided alternative.”



FIRST TAKE — one sentence before you work the parts

Would you compare each technician’s two times or treat them as separate people? Explain in one sentence.

[BOOK] SET UP A TEST FOR A MEAN (keep on the page)

For a population mean with unknown population standard deviation, use a one-sample t -test. Matched pairs are two linked measurements: subtract in the same order and analyze one sample of differences. Write hypotheses about the population mean, not the sample mean. The null uses equality; the alternative uses $<$, $>$, or $\neq$, as determined by the research question and subtraction order. Check random sampling or assignment, $n \leq 0.10N$ when sampling without replacement, and shape. For $n < 30$, the values being analyzed should have no strong skewness or outliers.

Work (a)–(c).

DOK 1 (a) Define μ_d in context.

DOK 2 (b) Check the random-sample, 10 percent, and small-sample shape conditions in three short notes.

DOK 3 (c) In 3–4 sentences, choose a setup and justify it. State the hypotheses using the given subtraction order and reduction question, then explain why the 36 times are not 36 independent people.

Frame: The independent units are _____; preserving each pair changes the analysis by _____.

Frame: With the stated subtraction order, the claim corresponds to _____; the rejected setup instead answers _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored **E / P / I**.